



الأكاديمية السعودية الرقمية
SAUDI DIGITAL ACADEMY



Saudi Regulations & Decisions Data Hub



By:

Shouq Alotaibi
Norah Aldakhil
Yaqeen Alhalal

1. Executive Summary

Saudi Regulations & Decisions Data Hub (SRDDH) is a Data Engineering project designed to collect and organize Saudi laws, executive regulations, and official decisions from government sources into a unified, structured dataset, enabling users, such as lawyers, to track how regulations and decisions evolve over time.

The project addresses the challenge of finding and tracking regulatory documents across different sources and preserving their historical versions. The pipeline collects raw data, cleans and standardizes it, validates data quality, detects document changes using content hashing, flag changed documents and assigns an edit type (new / unchanged / updated / removed). The processed data is stored in a relational database with full version history.

2. Problem Statement

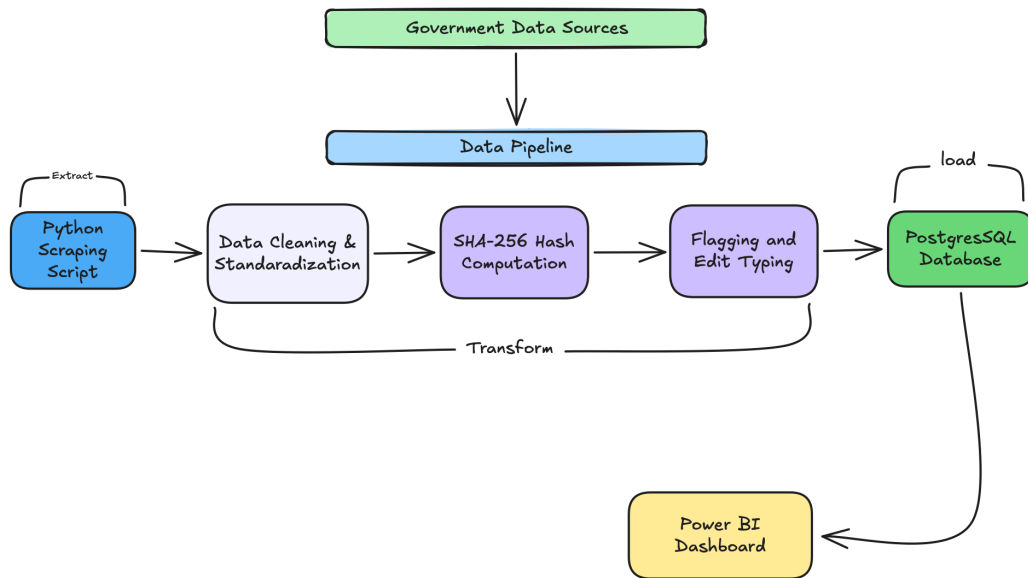
In Saudi Arabia, regulatory updates across governmental entities, such as the Ministry of Justice (MoJ), happen rapidly. However, government portals publish regulations as static HTML or PDF pages that overwrite prior versions when edits occur, lacking point-in-time version tracking and standardized metadata.

Pain Points

- **Manual & Error-Prone Tracking:** Legal and compliance teams spend hours manually searching and reading lengthy documents side-by-side to detect updates. This visual check process is highly inefficient and prone to human error.
- **Inconsistency Across Entities:** While select agencies like the MoJ occasionally note what has been changed or canceled, other governmental entity sites simply replace the HTML page without preserving metadata across historical versions.
- **Lack of Automated Version Tracking:** Even when a site marks a section as "changed," it usually displays only the new wording. It does not show side-by-side git-style files highlighting the exact words added, updated, or deleted over time.

3. Proposed Solution

The proposed solution implements a streamlined data platform designed to efficiently harvest, validate, structure, and visualize official legislative documents.



4. Project Objectives and End-Users

1. Automatically gather regulations and decisions from trusted government portals.
2. Store regulatory documents in a structured, consistent database format.
3. Maintain full point-in-time version history without overwriting legacy records.
4. Automatically detect and classify changes (new, updated, unchanged, or removed).
5. Validate data quality of incoming data.
6. Allow users to compare document versions side-by-side to highlight exact text edits.
7. Deliver a user-friendly dashboard to search, review, and track regulatory updates.

Dashboard Concept

If time permits, a Power BI dashboard will be developed to search and explore Saudi regulations and decisions.

Key Features

- **Search & Filter:** Search documents by title, issuing authority, or category.
- **Version History:** View all recorded versions of a document with collection dates.
- **Side-by-Side Comparison:** Compare older and newer versions to highlight text changes.

User Workflow Search Document → Select History → Compare Changes

Target End-Users

- **Legal & Compliance Officers:** To monitor regulatory risk and audit past compliance postures.
- **Human Resources & Legal Ops:** To keep corporate policies compliant with updated Saudization rules and labor laws.
- **Data Engineers & AI Practitioners:** To leverage clean, version-controlled legal datasets for training LLMs and RAG search indexers.

5. Data Sources & Data Fields

Primary Source (MVP)

- **Bureau of Experts at the Council of Ministers:** Official authority responsible for drafting and publishing Saudi royal decrees, laws, and regulatory decisions.

Extended Sources

- **Umm Al-Qura Official Gazette:** Historical publication registry.
- **Ministry of Commerce & SDAIA:** Executive mandates, corporate governance laws, and AI regulatory policies.

Field	Description
document_id	Unique identifier for each document
title	Title of the law, regulation, or decision
law_name_en	English name of the document, when available
authority	Issuing government authority, when available
issue_date	Official date the document was issued
publication_date	Official publication date of the document
status	Current status of the document, such as active
category	Document type: Law / Executive Regulation / Decision
source_url	URL of the original source
source_name	The name of the source
content	Full text of the document
content_hash	SHA-256 hash used to detect content changes
version_number	Version number assigned by the pipeline
fetches_at	Date and time when the document was collected
edit_flag	Flags changed documents
edit_date	The date when the document was edited.
edit_type	The type of the edit (new / unchanged / updated / removed)

6. Technical Stack

- **Git & GitHub:** Track code changes, manage feature branches, and collaborate.
- **Python:** Execute web scraping, text processing, hash generation, and ETL workflows.
- **Docker:** Package and isolate PostgreSQL, Airflow, and local development services.
- **PostgreSQL:** Relational database to store normalized documents, version history, and metadata.
- **Apache Airflow:** Workflow orchestrator to schedule, automate, and monitor daily ETL ingestion.
- **Microsoft Azure or Amazon Web Services:** Cloud platform for hosting managed PostgreSQL and live deployment.
- **Power BI / Streamlit:** Develop the interactive search and version-comparison dashboard.

7. Project Timeline and Key Deliverables

Week	Phase	Key Deliverables	Team Member
Week 1 (Aug 30th, 2026 - Sep 3rd, 2026)	Understand the Problem.	☐ Project Proposal Document. ☐ Project Proposal Presentation.	All members.
Week 2 (Sep 6th, 2026 - Sep 10th, 2026)	Design Database Models and Data Pipelines	☐ Database model design. ☐ Pipeline Architecture Diagram.	All members
Week 3 (Sep 13th, 2026 - Sep 17th, 2026)	Build the ETL Workflow	☐ Python web scraping, required data transformation steps, and loading to the destination script. ☐ Apache Airflow DAGs Setup.	All members.
Week 4 (Sep 20th, 2026 - Sep 24th, 2026)	Test & Evaluate	☐ End-to-End Pipeline Validation and Data Anomaly Handling. ☐ AWS Setup.	All members
Week 5 (Sep 27th, 2026 - Oct 1st, 2026)	Deploy & Present	☐ GitHub Repository and Final Project Presentation. ☐ An interactive dashboard	All members

8. Scope Boundaries

- Collect data from one official source: Bureau of Experts (laws.boe.gov.sa)
- Save the raw data with information about each collection run
- Clean Arabic text, fix dates, and standardize categories and authorities
- Check data quality and separate bad records
- Detect changes using hashing (new / unchanged / updated / removed)
- Save all old versions in PostgreSQL
- Compare two versions line by line
- Run the pipeline automatically every day with Apache Airflow
- Streamlit dashboard for search and version review

Out-of-Scope (Future Work): Extra data sources such as MoJ, OCR parsing for scanned PDFs, LLM/RAG summarization features, and user authentication systems.

9. Evaluation Plan

Criteria	Verification
Extraction Efficiency	Verify scraping scripts reliably capture complete text rows without missing fields.
Data Integrity & Versioning	Confirm PostgreSQL retains past document versions without overwriting existing rows.
Comparison Accuracy	Test the dashboard to ensure users can search seamlessly and review line-by-line text differences.

10. Risks & Mitigation

Identified Risk	Impact	Mitigation Strategy
Source Structure Changes	HTML changes break web extraction.	Modularize scrapers and monitor extraction logs.
Missing Data Fields	Required attributes unavailable or empty.	Apply strict validation before loading.
Duplicate Documents	Unchanged records re-ingested repeatedly.	Use SHA-256 hashes to bypass duplicate inserts.
Regulation Overwrites	Regulation Overwrites	Regulation Overwrites

11. Expected Outcomes

The implementation of this project delivers significant practical benefits:

- **Time & Effort Savings:** Replacing manual and tedious reading across scattered government portals with quick, automated data gathering and comparisons.
- **Enhanced Transparency:** Providing legal researchers, enterprises, and everyday users with clear and reliable visibility into how and when specific regulations change over time.
- **Solid Foundation:** Establishing a practical and scalable technical foundation that can easily be expanded to cover other government sectors or integrated into broader compliance systems in future phases

12. Conclusion

In conclusion, SRDDH transforms fragmented government portals into automated, audit-ready structured data. By automating the process of collecting documents and change detection, the pipeline continuously captures and archives legal updates, making regulatory changes easier to identify and review. This creates an organized and useful regulatory, version-controlled dataset ready to power downstream business analytics and enterprise AI while reducing manual effort.